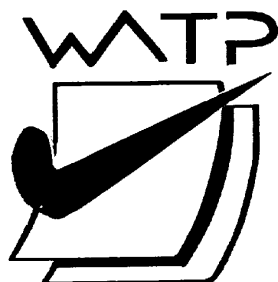


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SEMESTER TWO

**MATHEMATICS
METHODS
UNITS 1 & 2**

2018

SOLUTIONS

Calculator-free Solutions

1. (a) $\frac{2^{3x}}{2^y} = 2^4 = 16$ ✓✓
- (b) $u^{\frac{7}{2} - (\frac{5}{2} + \frac{1}{2})}$ ✓
- $= u^{\frac{1}{2}} = \sqrt{u}$ ✓ [4]
2. $ab = 15$ and $a + b = 8$
 $a = 3, b = 5$ or $a = 5, b = 3$ ✓
 $abx^2 + (2b + 7a)x + 14 = 15x^2 + cx + 14$
 $2b + 7a = 2(5) + 7(3)$ or $2b + 7a = 2(3) + 7(5)$ ✓
 $c = 31$ or 41 ✓✓ [4]
3. (a) $m = \frac{4}{3}$ ✓
 $-5 = \frac{4}{3}(3) + c$
 $\therefore c = -9 \therefore y = \frac{4}{3}x - 9$ ✓
 $-4x + 3y = -27$ ✓
- (b) $-8x + 6y = -14$
 $9x - 6y = 12$
 $x = -2 \quad y = -5$ ✓
 $D(-2, -5)$ ✓
- (c) $4(k - 2) - 3(2k - 3) = 7$ ✓
 $-2k + 1 = 7$
 $k = -3$ ✓ [7]
4. $\frac{1.2 \times 10^{-4}}{3 \times 10^{-7}} = 4 \times 10^2 = 400$ ✓✓ [2]
5. (a) $2(3xy) + 2(3x^2) + 2xy = 32$ ✓
 $3xy + 3x^2 + xy = 16$ ✓
 $3x^2 + 4xy = 16$
- (b) $V = 3x^2y$ and $y = \frac{16 - 3x^2}{4x}$ ✓
 $V = 3x^2 \left(\frac{16}{4x} - \frac{3x^2}{4x} \right)$ ✓
 $V = 12x - \frac{9x^3}{4}$
- (c) $V' = 12 - \frac{27x^2}{4}$
 $12 - \frac{27x^2}{4} = 0$ for stationary point ✓
 $x^2 = \frac{16}{9}$ ✓
 $x = \frac{4}{3}$ (discard $-\frac{4}{3}$) ✓

(d)

$V(x)$	$\uparrow$	$\frac{4}{3}$	$\downarrow$
$V'(x)$	$+$	0	$-$

 V has a maximum value

✓✓

[9]

6. (a) $T_{n+1} = T_n + 10$ $T_1 = -3$

✓✓

(b) $T_2 = -8$

✓

$T_3 = -32$

✓

[4]

7. (a) $y = 10x^2 - 2x^3 - 16x + c$
 $3 = 10(2)^2 - 2(2)^3 - 16(2) + c$
 $c = 11$

✓

✓

✓

$y = 10x^2 - 2x^3 - 16x + 11$

(b) Gradient of x -axis = 0

$\frac{dy}{dx} = 20(2) - 6(2)^2 - 16 = 0$

✓

 $\therefore m = 0$ Tangent is horizontal and parallel to the x -axis.

✓

[5]

8. (a) Vertical translation 12 units down

✓

Horizontal dilation by factor $\frac{1}{2}$.

✓

(b) $7^x = 7^{2x} - 12$

Let 7^x be k . $k^2 - k - 12 = 0$

✓

$(k - 4)(k + 3) = 0$

✓

$7^x = 4$ $7^x \neq -3$

✓

One solution, therefore intersects at only one point.

[5]

9. (a) $y = \frac{x^3}{2} + 2x - 7$

✓

$\frac{dy}{dx} = \frac{3}{2}x^2 + 2$

✓

$m = 8$

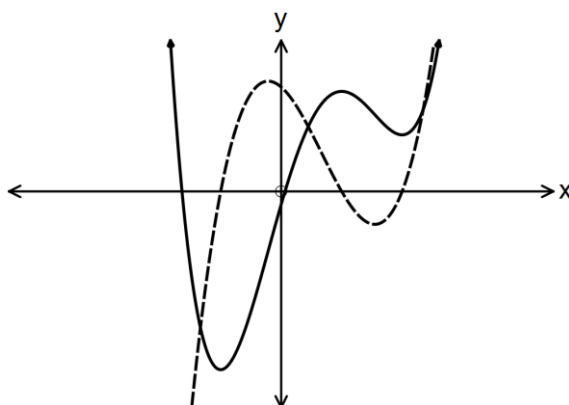
✓

$1 = 8(2) + c$

$y = 8x - 15$

✓

(b)

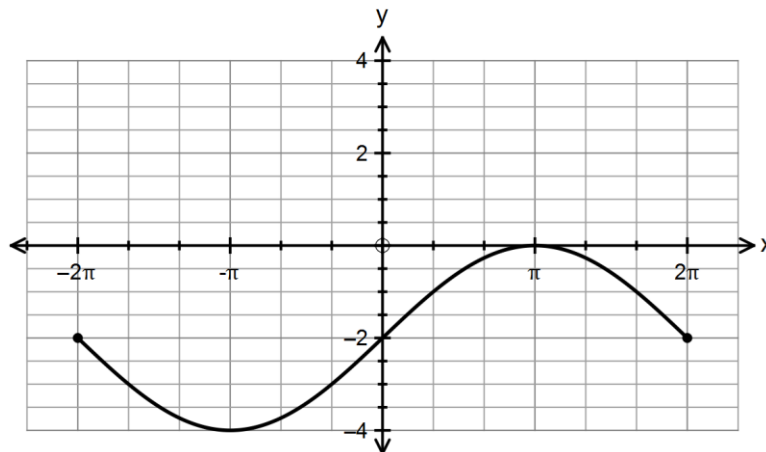


✓✓

[6]

10. (a) Period = 4π
Amplitude = 2
(b)

✓
✓



✓✓✓

[5]

Calculator-assumed Solutions

11. (a) The number of phones she is given to repair for the week.
(b) She fixes 23 per day, for 4 days $23 \times 4 = 92$ phones
(c) $\frac{108}{23} = 4.69565$ days
 $0.69565 \times 8 = 5.5652$ hours
5 hours and 34 minutes
(d) $T_1 = 155$ $T_2 = 128$ $T_3 = 99$ $T_4 = 68$
 $T_{n+1} = T_n - (25 + 2n)$ $T_0 = 180$

✓
✓
✓

✓

✓✓

[6]

12. (a) $x^2 + y^2 = 4$
(b) $\tan \theta = \sqrt{3} \therefore \theta = \frac{\pi}{3}$ radians
radius = 2 units
(c) $OB = \frac{2}{\cos \frac{\pi}{3}} = 4$

✓
✓
✓

Therefore area of $\triangle AOB = \frac{1}{2}(4)\sqrt{3} = 2\sqrt{3}$ units²

✓

$$\begin{aligned} \text{Area of sector} &= \frac{1}{2}r^2\theta = \frac{1}{2}(2)^2\left(\frac{\pi}{3}\right) \\ &= \frac{2\pi}{3} \text{ units}^2 \end{aligned}$$

✓

Area of shaded part = triangle – sector

$$= 2\sqrt{3} - \frac{2\pi}{3} = 1.37 \text{ units}^2 \text{ (3 sig fig)}$$

✓

[6]

13. (a) 5.1 seconds ✓
 (b) $v(t) = -9.8t + 25$ ✓✓
 $v(0) = 25$
 (c) Maximum height is 31.89 m when $t = 2.55$ sec
 $\frac{31.89 \times 2}{2.55 \times 2} = 12.5$ m/s ✓✓ [5]

14. (a) $f(x) = (2 + x)^2 = 4 + 4x + x^2$ ✓
 $\lim_{h \rightarrow 0} \frac{(2 + x + h)^2 - (2 + x)^2}{h} = 2x + 4$ ✓
 (b) (i) -7.5 ✓
 (ii) 18 ✓
 (c) $p'(x) = 3x^2 - 3a$ ✓
 $0 = 3(\sqrt{2})^2 - 3a$ ✓
 $a = 2$ ✓
 $-\sqrt{2} = (\sqrt{2})^3 - 3(2)(\sqrt{2}) + b$ ✓
 $b = 3\sqrt{2}$ ✓ [7]

15. (a) $3x^2 - \frac{4}{x^2} - 11 = 0$
 $x = -2$ or $x = 2$ ✓✓

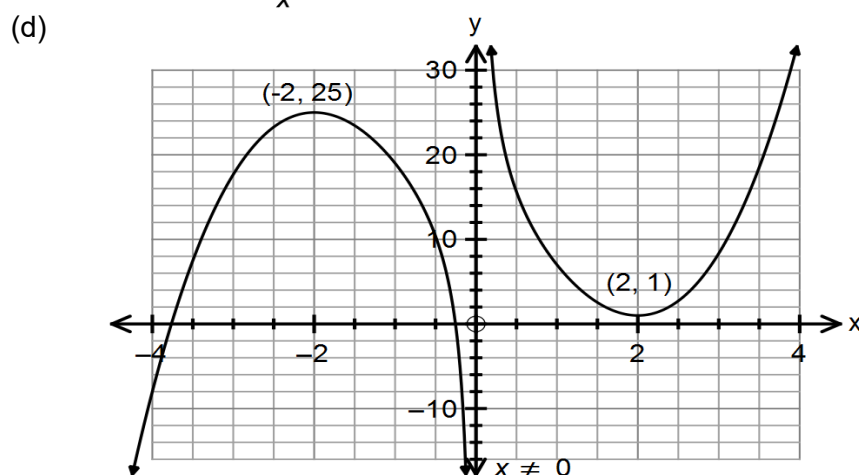
(b)

		-2		2	
y'	+	0	-	0	+
y	↑	-	↓	-	↑

$x = -2$ Maximum

$x = 2$ Minimum

- (c) $y = x^3 - 11x + \frac{4}{x} + c$ ✓
 $c = 13$ ✓
 $y = x^3 - 11x + \frac{4}{x} + 13$ ✓



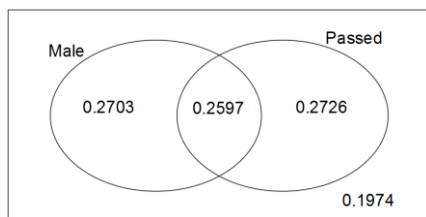
✓✓✓

[10]

16. (a) (i) $T_{100} = 23 + (99)(9)$ ✓
 $= 914$ ✓
(ii) $357980 = \frac{n}{2}(23 + 2534)$ ✓
 $n = 280$ ✓
(b) (i) $T_{n+1} = \frac{1}{4}T_n$ $T_1 = 12$ ✓✓
(ii) $S_{\infty} = \frac{12}{1 - \frac{1}{4}} = 16$ ✓✓ [8]
17. (a) 92°C (initial temp of tea) ✓
 22°C (room temp) ✓
(b) After 4.12 mins and before 7.45 mins ✓✓
 $4.12 \leq t \leq 7.45$
(c) Horizontal asymptote $y = 22$ ✓
The tea will cool at a decreasing rate as it approaches room temperature which is 22° . ✓ [6]
18. (a) $y = \frac{2}{x-3} + 2$ ✓✓
(b) $y = -3\sqrt{x+4}$ ✓✓
(c) $y = \left(\frac{1}{2}\right)^x - 4$ ✓✓ [6]
19. (a) 0.2 ✓
(b) 0.5 ✓
(c) 0.3 ✓
(d) $\Pr(X \cap Y) = \Pr(X) \times \Pr(Y)$ if independent ✓
 $\Pr(X \cup Y) = \Pr(X) + \Pr(Y) - \Pr(X \cap Y)$ ✓
 $\Pr(X) + \Pr(Y) - \Pr(X \cup Y) = \Pr(X) \times \Pr(Y)$ ✓
Let $\Pr(Y) = k$
 $0.5 + k - 0.8 = 0.5k$
 $k = 0.6 = \Pr(Y)$ ✓ [6]
20. (a) 1, 3, 7, 15, ...
 $T_{n+1} = T_n + 2^n$ $T_1 = 1$ ✓✓
(b) 20 sets ✓ [3]

21. (a) (i) $y = 16 - 6 - w$ ✓
 $y = 10 - w$
(ii) $y^2 = 36 + w^2 - 12w \cos Y$ ✓
(iii) $(10 - w)^2 = 36 + w^2 - 12w \cos Y$ ✓
 $64 - 20w = -12w \cos Y$ ✓
 $\therefore \cos Y = \frac{5w - 16}{3w}$
- (b) (i) $A = \frac{6w \sin Y}{2} = 3w \sin Y$ ✓
 $A^2 = 9w^2 \sin^2 Y$ ✓
(ii) $9w^2 \sin^2 Y = 9w^2(1 - \cos^2 Y)$ ✓
 $A^2 = 9w^2 \left(1 - \left(\frac{5w - 16}{3w} \right)^2 \right)$ ✓
 $A^2 = -16w^2 + 160w - 256$ ✓
- (c) (i) $A = \sqrt{-16w^2 + 160w - 256}$
 $A' = 0$ when $w = 5$ ✓
Maximum area = 12 units² ✓
(ii) $y = 10 - w = 5$
The triangle is isosceles. ✓ [12]
22. $y' = 3x^2 - 12x + k$ ✓
 $b^2 - 4ac = 0$ for one solution
 $144 - 4(3)k = 0$ ✓
 $k = 12$ ✓ [3]
23. (a) $W = W_0(1.085)^t$
 $R = R_0(0.95)^t$
 $10W_0 = R_0$
 $W_0(1.085)^t = 10 \times W_0(0.95)^t$ ✓
 $(1.085)^t = 10(0.95)^t$
 $t = 17.329$ years ✓
After 18 years there will be more wallabies. ✓
- (b) $W_{n+1} = 1.085 W_n$ $W_0 = 655$ ✓
 $W_5 = 985$ ✓ [5]
24. (a) $r = -\frac{3}{2}$ ✓
 $\frac{-2 \left(1 - \left(-\frac{3}{2} \right)^n \right)}{1 - \left(-\frac{3}{2} \right)} = 30$ ✓
 $n = 9.003$
 $\therefore 10$ terms ✓
- (b) $T_5 = a + 4d$ and $T_7 = a + 6d$
 $\therefore 2a + 10d = 38$ (eq 1) ✓
 $S_{15} = 375 = \frac{15}{2}(2a + 14d)$ (eq 2) ✓
 $a = 4$ $d = 3$ ✓
 $S_{30} = 15(2(4) + 29(3)) = 1425$
Sum of next 15 terms = $S_{30} - S_{15} = 1050$ ✓ [7]

25. (a)



- (b) (i) 0.5323
 (ii) 0.2726
 (iii) 0.51
 (iv) 0.8026

✓✓✓
 ✓
 ✓
 ✓
 ✓

[7]

26. The particle's initial displacement is 5 m to the right of the origin.

✓

$$v = 3t^2 - 12t \therefore \text{Initial velocity} = 0$$

✓

[2]